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Earth Observations and Machine Learning for Gridded Macroeconomic Data

Marco Marini, Jim Tebrake, and Andinet Woldemichael

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WORKING PAPER

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Statistics Department

Earth Observations and Machine Learning for Gridded Macroeconomic Data

Prepared by Marco Marini, Jim Tebrake, and Andinet Woldemichael*

Authorized for distribution by Jim Tebrake
September 2026

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ABSTRACT: This paper presents a methodology for downscaling official national and subnational macroeconomic data into high-resolution grids using spatial machine learning techniques. Traditional macroeconomic data are highly aggregated and obscure the spatial distributions needed to understand and quantify local economic activity, impacts of physical risks, and infrastructure gaps. Our hierarchical approach integrates official macroeconomic accounts with Earth observation predictors, such as nighttime lights, built-up areas, land cover, and gridded population, to deliver high-resolution, gridded estimates that remain fully consistent with official subnational and national totals. The empirical application focuses on constructing experimental gridded GDP by ten-sector industry for Canada and the United States. The methodology complements and enhances official statistics by adding spatial granularity through open geospatial datasets and can be extended to generate gridded capital stock estimates.

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WORKING PAPER

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I. Introduction

Economic activity is inherently spatial. Geography is an important determinant of economic activity, productivity, and income differences across space (Hall and Jones, 1999, Sachs, and others; Nordhaus, 2006). Production, consumption, investment, trade, infrastructure, and physical assets are distributed unevenly across geographic space, reflecting differences in population, natural resources, institutions, and connectivity. Studies also show that much of trade that involves the movement of final and intermediate goods is determined by location (Allen and Arkolakis, 2014). Exposure to physical risks, such as floods, hurricanes, and wildfires, is likewise location specific. As a result, geography plays an important role in shaping economic outcomes, productivity, welfare, and trade patterns.

Despite the importance of location, most macroeconomic models and official macroeconomic statistics remain highly aggregated. Traditional policy analysis often relies on representative agents, national averages, and calibrated parameters that abstract from spatial heterogeneity (Aligishiev et al., 2023)[1]. While recent advances in heterogeneous-agent and spatial general equilibrium models increasingly incorporate geographic dimensions (Dvorkin, 2023; Auclert et al. 2025; Allen and Arkolakis, 2014), their application is frequently constrained by the lack of spatially detailed and internally consistent macroeconomic data.

Official GDP statistics illustrate this limitation. Although they are compiled from activities occurring across many locations, the aggregation process largely removes information on the geographic distribution of economic activity. Some advanced statistical systems publish subnational GDP estimates, such as county-level GDP in the United States, provincial GDP in Canada, and regional GDP within the European Union. However, even these statistics are often too coarse to support analyses requiring fine spatial detail.

To address this gap, this paper develops a methodology for producing gridded GDP estimates by combining Earth observation data, spatial machine learning, and statistical reconciliation with official national accounts. The framework uses high-resolution geospatial indicators, including population, nighttime lights, building footprints, and land-cover data, to downscale official GDP into 1 km² grid cells while preserving consistency with national and subnational totals. The resulting gridded estimates provide a foundation for analyzing localized economic activity, climate and disaster risks, infrastructure gaps, and regional disparities, particularly in settings where granular official data are limited or unavailable.

The contribution of the paper to the literature is twofold. First, it introduces the industry dimension at the grid level, providing additional information on the share of industry. Second, it proposes a statistical reconciliation approach to ensure that the grid cell level GDP estimates and industry shares are reconciled with official data at different levels of aggregation consistent with the System of National Accounts (SNA).

The rest of the paper is organized as follows. Section II reviews the relevant literature. Section III illustrates the data sources used for the application to Canada and the United States. Section IV discusses methods and proposed indicators. In Section V, we discuss the results before concluding in Section VI.

[1] In the literature of physical risks and their impacts on the financial systems among other sectors of the economy, open-source granular data-intensive risk models, such as CLIMADA, are example albeit restricted to partial or sectoral analysis [Håkansson et al., 2025; Eberenz et al., 2020].

II. Related Work

Geography is one of the most important factors determining divergence in economic growth and income across space (Nordhaus, 2006). In growth literature, geography—measured by distance from the coast, distance from the equator, climatic zones, etc. are found to be important exogenous determinants of divergence in incomes and growth (Hall and Jones, 1999, Sachs, and others). However, macroeconomic data are highly aggregated and often lack information on spatial distribution, which is also underscored in the literature that averages of most geographic variables such as temperatures across larger administrative regions tend to be meaningless. He proposed spatial rescaling of aggregate economic data to grid cells by "inferring the distribution of data in one set of spatial aggregates based on the distribution of another set of spatial aggregates" as in Mendelsohn et al. (1994). In this regard, Nordhaus (2002) is one of the early studies to rescale aggregate economic output into spatial grids using a range of spatial rescaling approaches including area-weighted averages, and constrained smoothing that imposes smoothness for spatially correlated cases.

Nordhaus (2006) coined the term Gross Cell Product (GCP) to capture economic activity at the grid-cell level, with the same conceptual interpretation as GDP. The difference is that instead of a country or another administrative region, GCP represents the output of a grid cell at a specified resolution. The author published the first gridded global economic activity data called "G-Econ" at 1x1 degree resolution grid cells[2]. Subsequently, Nordhaus and Chen (2006, 2008) produced disaggregated GDP data for all large countries at 1x1 degree resolution.

In recent years, the methods of recovering spatial distribution of aggregate data have become significantly improved, especially with advancement of statistical and machine learning techniques as well as availability of high-resolution socioeconomic data, such as gridded population, urban settlement, built-up areas, and nighttime lights. The literature could be split into two strands along the level of resolution and treatment of political boundaries.

The first strand of the literature focuses on high-resolution gridded economic data, primarily GDP below the lowest subnational boundaries (for example, Nordhaus, 2006). Subsequently, the literature grew into mainstream economics and multidisciplinary subjects such as climate change. In a recent paper, Rossi-Hansberg and Zhang (2025) applied a machine learning model to estimate local GDP.[3] The second strand of literature focuses on downscaling national GDP to subnational political administrative boundaries, typically administrative levels 1 or 2, as opposed to grids. In their study of determinants of regional development, Gennaioli et al (2013) constructed subnational GDP from 1,569 subnational regions in 110 countries. Wenz et al. (2020) harmonized reported economic output data.[4] The dataset covers the period between 1960 and 2020. Recently, Kummu et al. (2025) also published a similar study that downscales GDP per capita PPP to administrative level 2 for 1990-2022 globally. Further, gridded GDP and economic activity has been widely applied in the climate projection literature under the Intergovernmental Panel on Climate Change (IPCC) Shared Socioeconomic Pathways (SSPs) which describe future socioeconomic conditions under different

[2] At the equator, a 1x1 degree resolution is roughly 111 km × 111 km.

[3] The dataset is published here: Local GDP Estimates Around the World | Becker Friedman Institute.

[4] The database is called "Database of Subnational Economic Output (DOSE)". Wenz et al. (2020) harmonized the data using information form statistical agencies, yearbooks, and other official publications—mainly EuroStat, OECD Regional Statistics, National Statistical Institutes, UND Human Development Reports—to harmonize reported data from 1,661 subnational regions in 83 countries.

climate scenarios including Jones and O'Neill (2016), Murakami and Yamagata (2019), Murakami et al. (2021), Wang and Sun (2022), and Jing et al. (2022).

In addition to gridded GDP, satellite imagery has been used to proxy economic activities at a granular level. For instance, Henderson et al. (2012) first showed that nighttime light is highly correlated with the level of local economic activities and hence can be a good proxy for wealth, income, and GDP at the macro level.[5] Other satellite imagery such as Normalized Difference in Vegetation Index (NDVI), which measures the level of greenness, are used to estimate agricultural yield and predict drought. Closely related, Ru et al. (2023) disaggregate national and subnational agricultural GDP into a global gridded dataset at roughly 10×10 km using satellite indicators of crop, livestock, fishery and forestry production. One of the advantages of these imageries is that they are made publicly available at a very high resolution, with frequent updates. Notable limitations include temporal inconsistencies between satellite sensors over a long period of time, underestimation in some sectors, and the lack of sectoral disaggregation to inform sector-specific policies and analysis.

Similarly, gridded population and built-up area are used in the literature as good proxies for gridded factors of production, labor and physical capital, respectively. Gridded Population of the World (GPW)[6] downscales national or subnational level population data into grids. The dataset had been subsequently revised over the years with the latest version providing population estimates for 2000, 2005, 2010, 2015, and 2020 at a 30 arc-second grid resolution. Similarly, WorldPop is another authoritative, open data source for gridded population and a wealth of other datasets including administrative areas, age and sex structures, dependency ratios, development indicators, and urban change.[7]

While these gridded data close some gaps in granularity of some macroeconomic data and statistics, there are some limitations and caveats. One of the limitations is that grid cell values are not straightforward to interpret without additional information on the type of economic activity. Disaggregating grid-level GDP values by industry—agriculture, manufacturing, services, etc.—could provide additional information that helps interpret total GCP values. This paper aims to provide these additional dimensions by downscaling GDP by industry shares.

III. Data Sources

The training data for our spatial machine-learning models combine official GDP and ten-sector industry statistics reported at national and subnational levels, which we downscale onto a high-resolution grid. The availability of official subnational GDP data varies considerably across countries. As a pilot, we consider the United States and Canada, which are data-rich countries. The U.S. BEA, for example, publishes GDP and

[5] The release of satellite imagery from NASA's Visible Infrared Imaging Radiometer Suite (VIIRS) since 2012, a modern satellite constellation, that replaced the DMSP-OLS and addressed some of the limitations and offer high-resolution imageries. In policy work, nighttime light is now one of the strong predictors of GDP growth and used extensively by the IMF for nowcasting economic activity. Similarly, Nitrogen dioxide (NO_2), which is the result of combustible fuel use such as in vehicles and machinery, has been a good proxy for economic activity at a local level particularly in places or countries where the use of such fuel use is prominent. NO2 data from 2005 to 2022 is available from the Ozone Monitoring Instrument (OMI), a Dutch-Finnish instrument, on the NASA Aura satellite.

[6] (GPW) is one of the many gridded socioeconomic datasets published by a group of demographers at NASA's Socioeconomic Data and Applications Center at Columbia University's Center for Integrated Earth System Information (SEDAC/CESIN). The first version (GPW v1) was published in 1995 with data on population counts and density at different levels of resolution globally. (https://www.earthdata.nasa.gov/data/projects/gpw)

[7] The full list of gridded datasets available from WorldPop can be accessed at https://www.worldpop.org/datacatalog/.

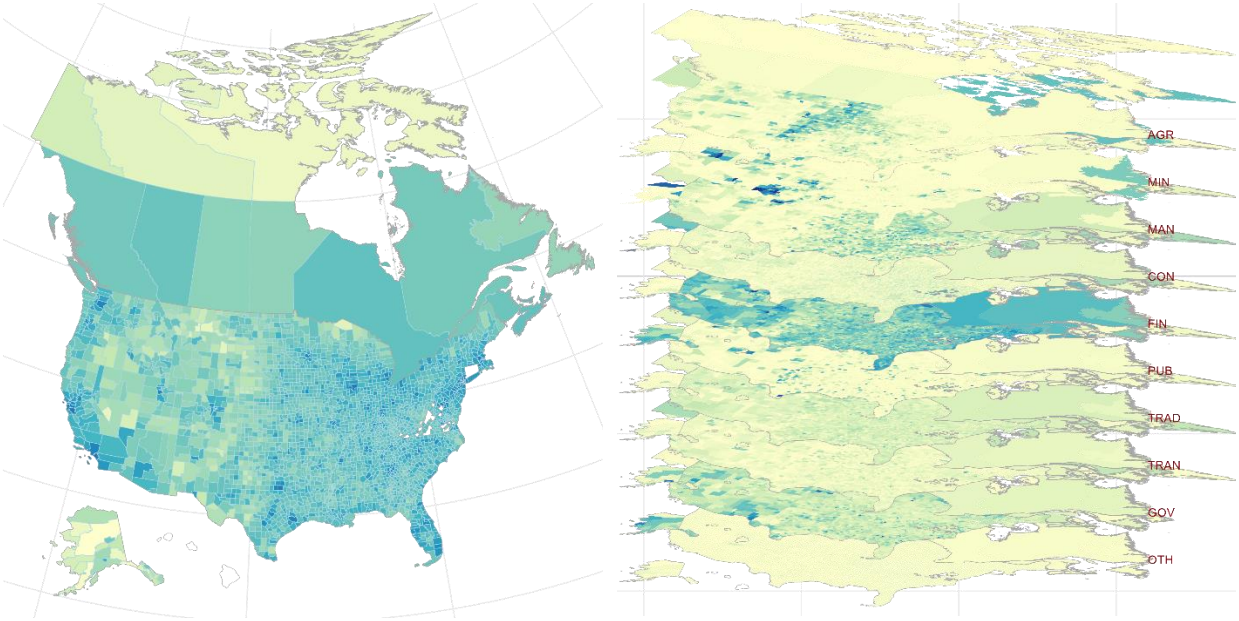
industry series at admin level 0 (country), admin level 1 (state), and admin level 2 (county); StatCan publishes comparable data at admin level 0 (country) and admin level 1 (province). We harmonize the available national and subnational official data using the ten-sector classification of the Groningen Growth and Development Centre (GGDC) 10-Sector Database (Timmer et al., 2014), which organizes economic activity under ISIC Rev. 3.1 into: agriculture, hunting, forestry and fishing; mining and quarrying; manufacturing; utilities (electricity, gas and water supply); construction; wholesale and retail trade, hotels and restaurants; transport, storage and communication; finance, insurance, real estate and business services; government services; and community, social and personal services (see Appendix 2).

Figure 1, panel (a) below shows a map of official subnational GDP data for Canada (provincial level) and the United States (county level) in 2021. We adjusted subnational GDP into internationally comparable units using the IMF World Economic Outlook (WEO) GDP in PPP international dollars as a benchmark. Figure 1, panel (b) shows the spatial distribution of ten-sector industry shares in 2021.

Figure 1: Official Subnational Real GDP and ten-sector industry shares, Canada and United States

(a) Log GDP per square km, real, Purchasing power parity (PPP) International dollar, 2021

(b) Ten-sector industry shares 2021



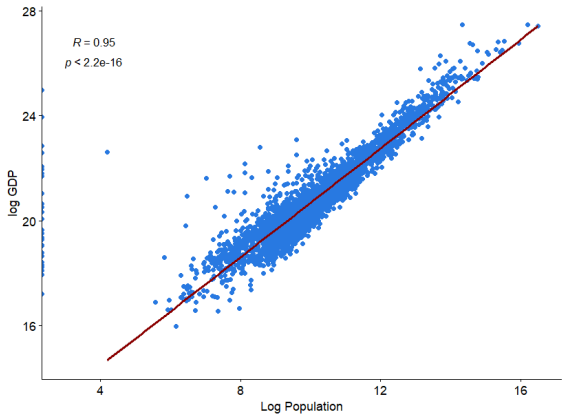
Source: Authors' computation based on data from BEA and StatCan.
Note: Ten-Sector industry are: AGR=Agriculture, MIN=Mining, MAN=Manufacturing, CON=Construction, FIN=Finance & Business Serv., PUB=Utilities, TRAD=Trade & Accom., TRANS=Transp. & Comm., GOV=Government & Pub. Admin, and OTH=Other Services. Data are county level for the U.S. and provincial level for Canada.

We use a set of high-resolution geospatial covariates to train the models at the lowest subnational level for which GDP by industry data is available. We use WorldPop's harmonized geospatial covariates which come at 3 arcsecond (~100m) resolution. Key sets of geospatial covariates we use include gridded population, nighttime lights, distances, building footprint, temperatures and precipitation, Nitrogen dioxide (NO_2), and number of iron and steel plants within 50 km grid. Population is a key determinant of economic activity as it measures labor and consumer demand. Nighttime light is an image of earth taken at night and is another important proxy of economic activity—places where there is more economic activity tend to be more lit at night. In countries where vehicles and machinery use combustible fuel, NO_2 emissions proxy the level of economic

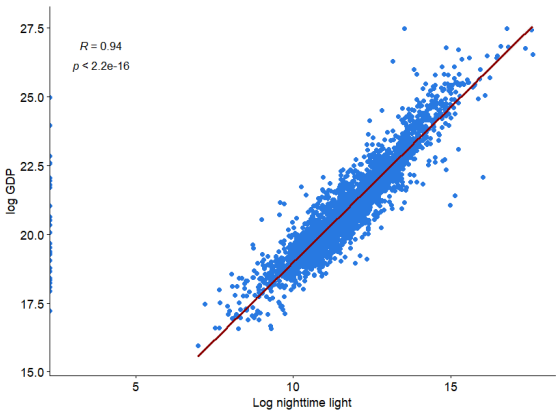
activities involving transportation of goods and people as well as production activities. We also use plant location data of heavy industries including iron and steel, cement, and oil and gas extraction from Global Energy Monitor, Global Iron and Steel Tracker (2025). As shown in Figure 2 panels (a) and (b), GDP is highly correlated with some of these covariates, for instance with population (R^2= 0.95) and nighttime light (R^2=0.94).

Figure 2: Scatter Plots of Subnational GDP vs (a) Population and (b) Nighttime Light: Canada and United States combined

(a) Scatter plot of log GDP and log population



(b) Scatter plot of log GDP and log nighttime light radiance



Source: Authors' computation.

Distance to built surface, distance to highway, and distance to cultivated land are another important set of covariates that provide information on the location of settlements and economic activities by sector. For instance, agriculture and extractive activities (forestry, fishing, mining and quarry) happen further away from urban areas with densely built surfaces, whereas services, such as financial services, typically happen in urban areas. Similarly, transportation sector is determined by availability of roads, railroads, harbors, airports, etc. Building footprint data on the number of buildings, average area, and density serves as another set of geospatial covariates to predict the volume of physical capital, concentration of economic activities, and infrastructure. Climatic variables such as temperatures and precipitation, and land cover are good proxies of agriculture and, to some extent, tourism activities. To this end, we compute the fraction of each grid cell covered by bare or sparsely vegetated land, built-up area, cropland, grassland, moss and lichen, permanent water bodies, shrubland, snow and ice, and tree cover, drawn from the ESA Earth Observation land cover product, which distinguishes eleven classes in total. Figure 3 shows maps of selected geospatial covariates used for baseline model training and prediction. Appendix A shows the complete list of high-resolution data used for training.

The spatial support of the training data differs materially across the two countries. U.S. models can draw on county-level outcomes, whereas Canadian models rely on 13 provincial and territorial observations. Accordingly, the Canadian results should be interpreted as experimental model-based spatial allocations, and not as independently validated estimates of subprovincial GDP.

Figure 3: Selected High-resolution Geospatial Covariates for Baseline Model Training and Prediction

(i) Log Population (2020) (ii) Log average Nighttime Light (2020) (iii) Crop land (2021)



Source: Authors' computation using data from WorldPop and ESA.

IV. Estimation Methods

We follow a two-step approach to downscale official subnational GDP and ten-sector industry shares to the grid level. In the first step, we train spatial random forest models for GDP and for each industry at the finest administrative level at which official data are available. To assess variation in performance, we train separate models on the pooled sample and on subsamples, apply variable selection within each, and retain the specification with the lowest root mean square error (RMSE). In the second step, we generate grid-level predictions from the selected models and their corresponding high-resolution geocovariates. We then reconcile these predictions in two directions. Vertical reconciliation constrains predicted grid values to sum to the official subnational totals. Horizontal reconciliation adjusts the predicted industry allocations so that the estimates are jointly consistent across geographic and industry dimensions. We describe each step in turn below.

IV. Machine Learning Estimates

Following the literature, we train predictive spatial machine learning models that consider spatial dependency (see Meyer et al., 2026; Rossi-Hansberg and Zhang, 2025 among others).

Let y_i^k denote the target variable observed at subnational administration level i and $k \in \{\text{logGDP}, \text{logGDP}_{\text{perKM}}, \text{AGR}, \text{CON}, \text{FIN}, \text{GOV}, \text{MAN}, \text{MIN}, \text{OTH}, \text{PUB}, \text{TRAD}, \text{TRAN}, \text{SER}\}$. And let $\boldsymbol{X}_i^k$ denote a vector of covariates aggregated to administrative level i. The covariates include population, nighttime light, NO_2, distances to nearest highway, cultivated land, built surface, average annual temperatures and precipitation, number and density of buildings, number of heavy industries within 50km radius, and land cover classification.

For each target variable k, the model is given by $y^k = f(\boldsymbol{X}^k) + \varepsilon^k$, where $f(\cdot)$ is unknown function and ε^k is an additive error term. The model is fitted using spatial random forest algorithm. For each response variable, we train the models using different pooled and split subsamples of geographic areas to see how model performance varies across regions with different morphology and socioeconomic features. The pooled sample

combines data from across the two countries.[8] As discussed below, model performance and predictive powers of covariates vary significantly across sample regions and responses (industries). For instance, while population and nighttime light strongly predict GDP, crop coverage of land cover tends to predict agricultural activity. Because models are estimated on covariates aggregated to the administrative unit and then applied at grid level, the estimated relationship is assumed to hold at cell scale.

We assess in-sample model performance using RMSE for out-of-sample predictive performances of the models. We follow the literature and use k-fold cross-validation (CV) strategy using leave-one-out method—an approach typically used in spatial machine learning literature with spatial dependence (see Meyer et al. 2026). Variables are selected using forward feature selection. We use CAST Version 1.04 library in R developed by Meyer et al. (2026).

We impose grid level constraints that ensure the sum of predicted shares of the ten sectors is one at the administrative level (state for the U.S., province for Canada). We also impose the condition that predicted industry shares are zero in areas where predicted GDP is zero such as in large permanent water bodies and uninhabited deserts and permanent snow.

IV.B Statistical Reconciliation

In the second step, we combine machine-learning (ML) predictions with official subnational statistics to produce estimates that are both granular and fully consistent with published subnational data. The starting point is the set of estimates for each grid cell and industry within a country derived in the first step. While these predictions capture rich spatial and sectoral patterns, they are not guaranteed to align with official statistics when aggregated to subnational entities or industries. The reconciliation step adjusts ("reconciles") these predictions so that they exactly match official totals, while preserving the underlying structure of the ML estimates.

To ensure cross-country comparability and robustness, we anchor the reconciliation to the first level of subnational data. This choice maintains consistency of the methodology across countries while preserving the granularity of the ML estimates, which could otherwise become uneven or fragmented due to differences in availability and quality of official data at more detailed administrative levels (e.g., admin 2).[9] To achieve this, we impose two types of consistency requirements. First, for each region and industry, the sum of grid-level estimates must equal the corresponding official regional total. Second, for each region, the sum across all industries must match the official regional aggregate. Together, these constraints ensure full alignment with the subnational accounting framework at the industry level.

The adjustment is carried out using a quadratic reconciliation approach following Di Fonzo and Marini (2011). Intuitively, we search for a new set of grid level industry estimates that are as close as possible to the original ML predictions, subject to the requirement that all official constraints are satisfied exactly. Closeness is defined in relative terms: deviations are penalized proportionally, so that areas or industries with larger initial values are not disproportionately adjusted. This preserves the spatial distribution and industry composition implied by the ML model. The resulting estimates have three key properties. First, they are fully consistent with official

[8] There are only 13 (province level) subnational GDP and industry data for Canada compared to 3,082 (county-level) for the United States. For Canada, estimation relies on 13 provincial observations (vs. 3,082 U.S. counties); consequently, no sub-provincial official data enter the estimation, and all within-province spatial detail is model-generated.

[9] While reconciliation is anchored at the first administrative level, data at the admin 2 level are still used in the estimation stage to derive the ML predictions through covariates. For example, we are using U.S. county-level official data to train the model for ML estimates in this paper.

regional and industry totals. Second, they preserve the predicted spatial distribution and sectoral variation. Third, the adjustment depends only on the discrepancy between aggregated ML predictions and official statistics, ensuring transparency in how corrections are applied.

From a computational perspective, the method is efficient and scalable. The underlying system exploits the sparse structure of spatial and industry aggregation, allowing it to be applied to high-resolution grids within each country. Further technical details on the statistical reconciliation method used are provided in Appendix B.

V. Application

This section presents the results of the two-step approach to derive gridded GDP estimates for Canada and the United States. Table 1 documents the spatial coverage of the gridded GDP dataset across the ten largest provincial and state economies in Canada and the United States. Coverage is measured by the number of 1 km² grid cells with positive post-benchmarking GDP, which reflects the presence of detectable economic activity in the Earth observation covariates used in the spatial random forest estimation. In Canada, Ontario accounts for the largest share of both cells and GDP, followed by Quebec and Alberta, with the three provinces together representing over 73 percent of total Canadian gridded GDP. British Columbia, despite its relatively modest cell count of 79,828, contributes 14 percent of national gridded GDP, reflecting the high economic density of the Greater Vancouver area. In the United States, California, Texas, and New York are the three largest economies by gridded GDP, collectively accounting for approximately 31 percent of the national total, consistent with their dominance in official national accounts. Detailed coverage statistics for all 13 Canadian provinces and territories and all 51 U.S. states and the District of Columbia are provided in Appendix C.

Table 1: Gridded GDP Coverage — Top 10 Provinces and States by GDP, 2021

A. Canada					B. United States				
Province	Cells	% Cells	GDP (USD PPP bn)	% GDP	State	Cells	% Cells	GDP (USD PPP bn)	% GDP
Ontario	190,495	19.3	820.5	38.2	California	197,269	3.7	3,504.1	14.8
Quebec	137,186	13.9	425.9	19.8	Texas	462,199	8.6	2,075.8	8.8
Alberta	216,265	21.9	333.1	15.5	New York	140,389	2.6	1,924.0	8.1
British Columbia	79,828	8.1	300.6	14.0	Florida	107,370	2.0	1,296.5	5.5
Saskatchewan	179,854	18.2	75.1	3.5	Illinois	171,827	3.2	949.2	4.0
Manitoba	82,059	8.3	66.7	3.1	Pennsylvania	132,115	2.5	848.3	3.6
Nova Scotia	33,071	3.3	42.4	2.0	Ohio	135,128	2.5	764.8	3.2
New Brunswick	35,835	3.6	33.9	1.6	Georgia	143,353	2.7	712.3	3.0
Newfoundland and Labrador	22,389	2.3	32.5	1.5	New Jersey	23,483	0.4	701.3	3.0
Prince Edward Island	6,588	0.7	7.1	0.3	Washington	108,233	2.0	701.1	3.0
Total Canada	**988,661**	**100.0**	**2,148.9**	**100.0**	**Total United States**	**5,354,938**	**100.0**	**23,681.2**	**100.0**

Notes: Grid cells reported are those with positive GDP only; cells with estimated no economic activity are excluded. Top 10 provinces/states ranked by post-benchmarking gridded GDP. Each grid cell covers approximately 1 km² at mid-latitude. GDP in PPP USD billions. % Cells and % GDP are shares of the respective national total. Full coverage for all provinces and states is in Appendix Table C-1.

Source: Authors' computation.

Table 2 shows typical model performance metrices—RMSE, R^2 and Mean Absolute Errors (MAE)—for the machine learning estimates. For GDP, we estimate four model variants: log GDP, log GDP per km^2, log GDP per capita, and log GDP per capita per km^2; for sector shares, we use shares as response variable. For the last two model variants we excluded population from the set of covariates. The RMSEs for log GDP and log GDP per km^2 are, respectively, 0.39 and 0.44, which are equivalent to an R-squared value of 0.95. This is comparable to model performance in similar work, such as Rossi-Hansberg and Zhang (2025) whose model's performance is 0.95 for GDP growth. The performance of log GDP per capita and log GDP per capita per km^2 are relatively lower, with a R^2 value of 0.5 and 0.56, respectively.

Table 2: Gridded GDP Model Performance: Canada and United States

Models	RMSE	R^2	MAE
log GDP	0.39	0.95	0.29
log GDP per km^2	0.44	0.94	0.32
Agriculture	0.06	0.54	0.04
Construction	0.03	0.17	0.02
Finance & Business Serv.	0.10	0.44	0.07
Government & Pub. Admin	0.08	0.04	0.06
Manufacturing	0.11	0.30	0.08
Mining	0.10	0.48	0.06
Other Services	0.01	0.19	0.08
Utilities	0.09	0.12	0.05
Trade & Accom.	0.05	0.19	0.04
Transp. & Comm.	0.04	0.18	0.02

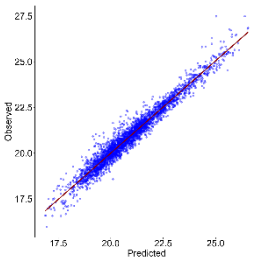
Source: Author's computations.

For the sectoral shares, model performance varies considerably, reflecting differences in the predictive power of the covariates across sectors. For instance, agriculture and mining sector models perform well, with R^2 values of 0.54 and 0.48, followed by Finance and Business Services and Manufacturing with values of 0.44 and 0.30, respectively. Other sectors tend to have lower model performance (e.g., 0.04 for the government sector). The variation in model performance across industries could be attributed to the extent to which the geographic features observed can predict spatial patterns in some sectors, particularly services. Figure 4 shows model performance across the response variables with x-axis showing predicted values and y-axis observed values.

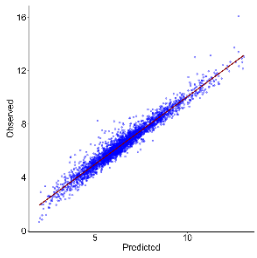
Variable selection is an important step in predictive machine learning to avoid overfitting. Figure 5 shows the set of covariates that yield the best model performance. The predictive power of the variables varies across response variables. For instance, built-up area, population, and nighttime light are among the top-ranking covariates selected to predict log GDP, whereas the share of cropland is the top variable that explains the model for agriculture and a few other sectors. It should be noted that rankings reflect each covariate's predictive contribution only; they are non-directional and imply no causal relationship with the sector.

Figure 4: Predicted vs. Observed GDP

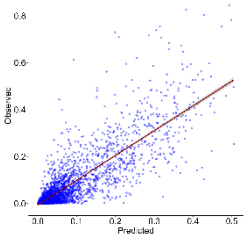
(i) Log GDP



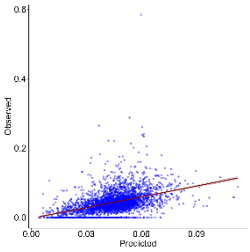
(ii) Log GDP per km



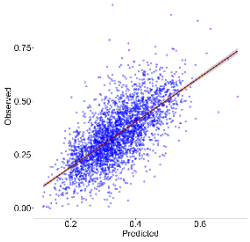
(iii) Agriculture



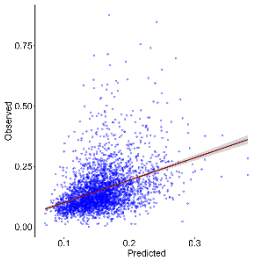
(iv) Construction



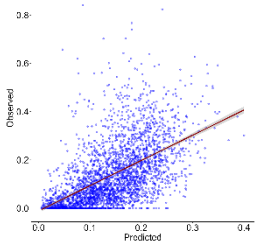
(v) Finance and Business Services



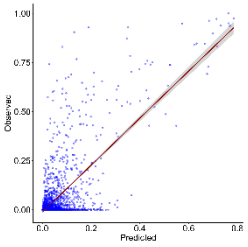
(vi) Government



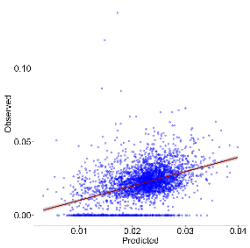
(vii) Manufacturing



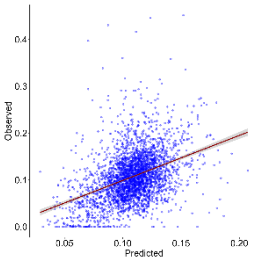
(viii) Mining



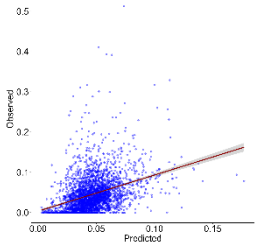
(ix) Other Services



(x) Trade and Communication



(xi) Transport and Communication



Source: Authors' computation.

Note: The chart for Utilities sector is not presented as the model predicts poorly.

Figure 5: Variable Selection by Response Variable

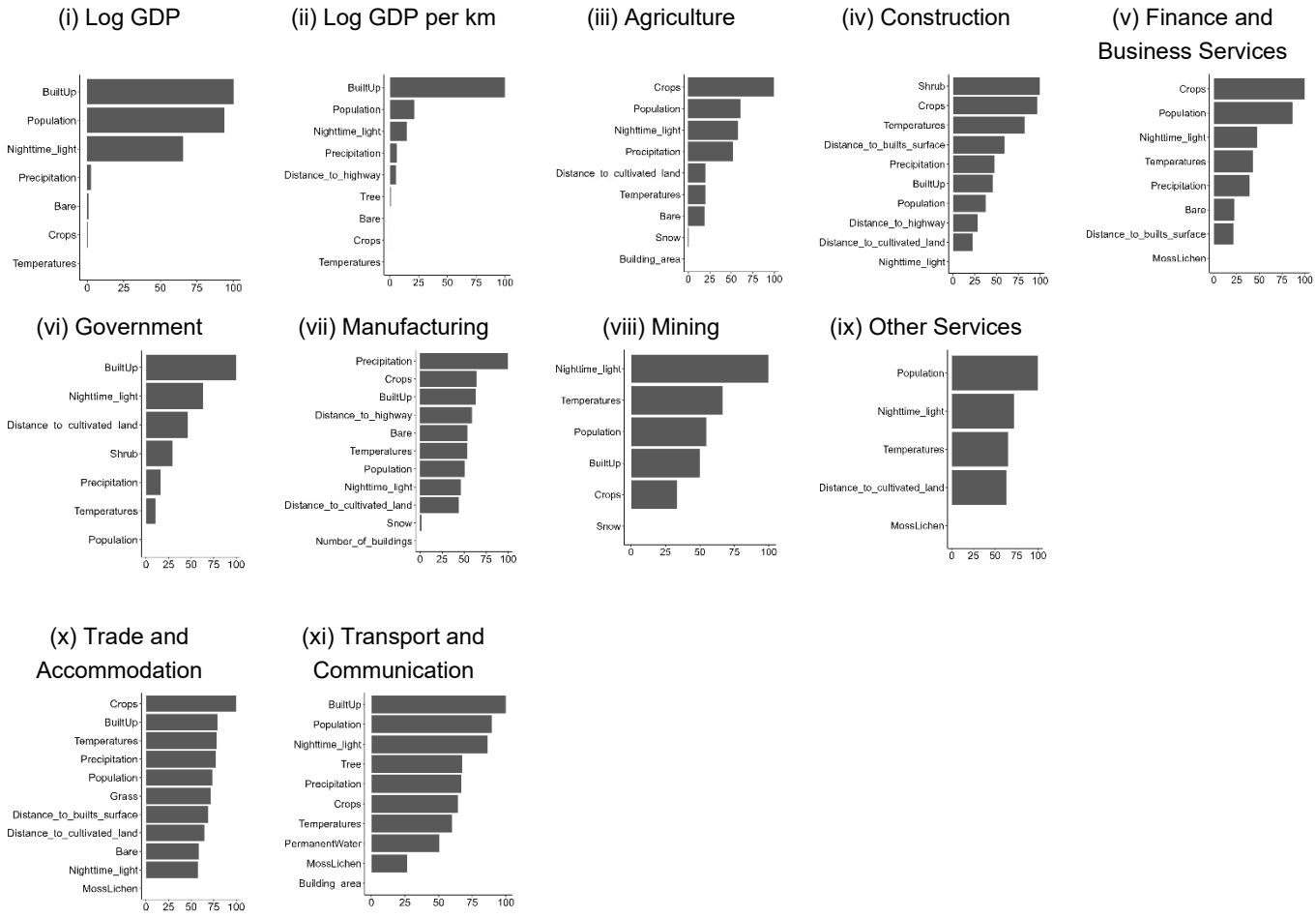
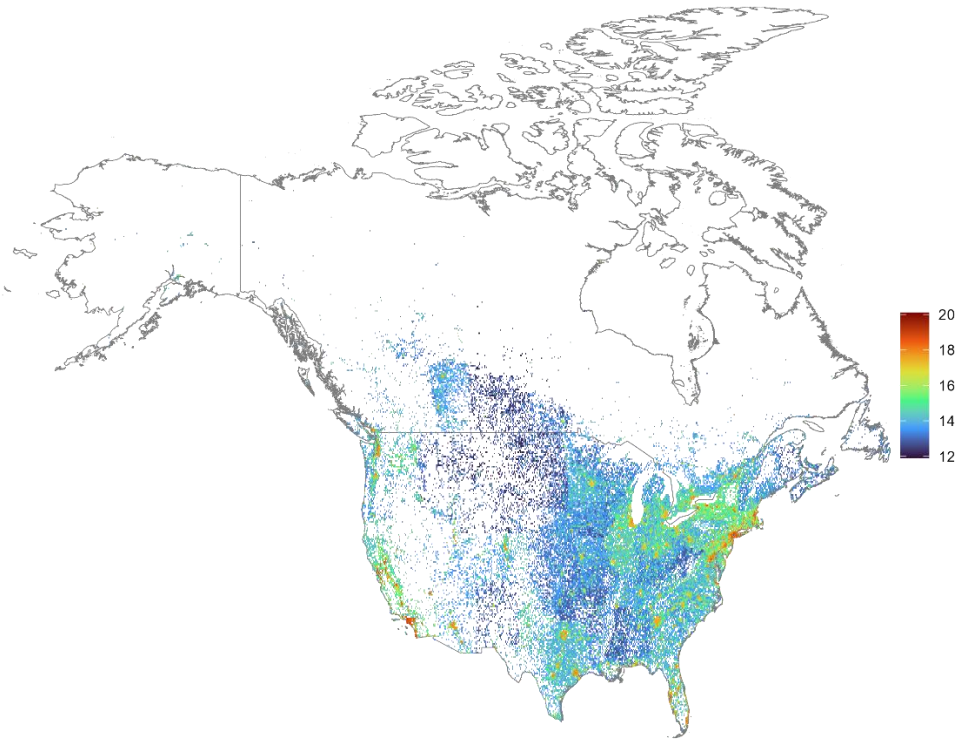


Figure 6 panels (a)-(c) show predicted gridded GDP and sectoral shares for agriculture and manufacturing at 1 km^2 resolution. The corresponding plots for predicted shares for all other sectors are shown in Appendix C.

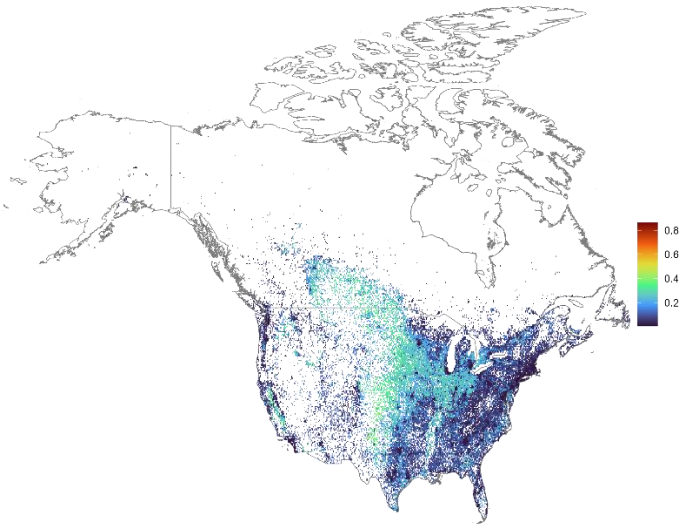
Note: The chart for Utilities sector is not presented as the model predicts poorly.

Figure 6: Predicted Gridded GDP and Ten-sector Shares, 1 km^2 resolution

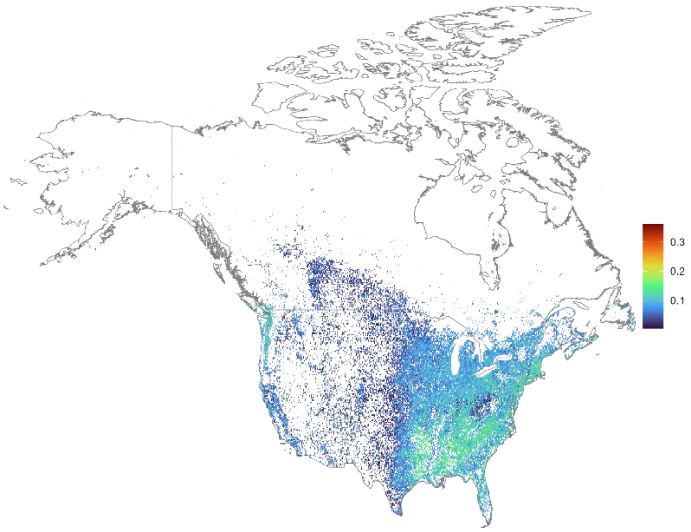
(a) Predicted log gridded GDP



(b) Share of agriculture



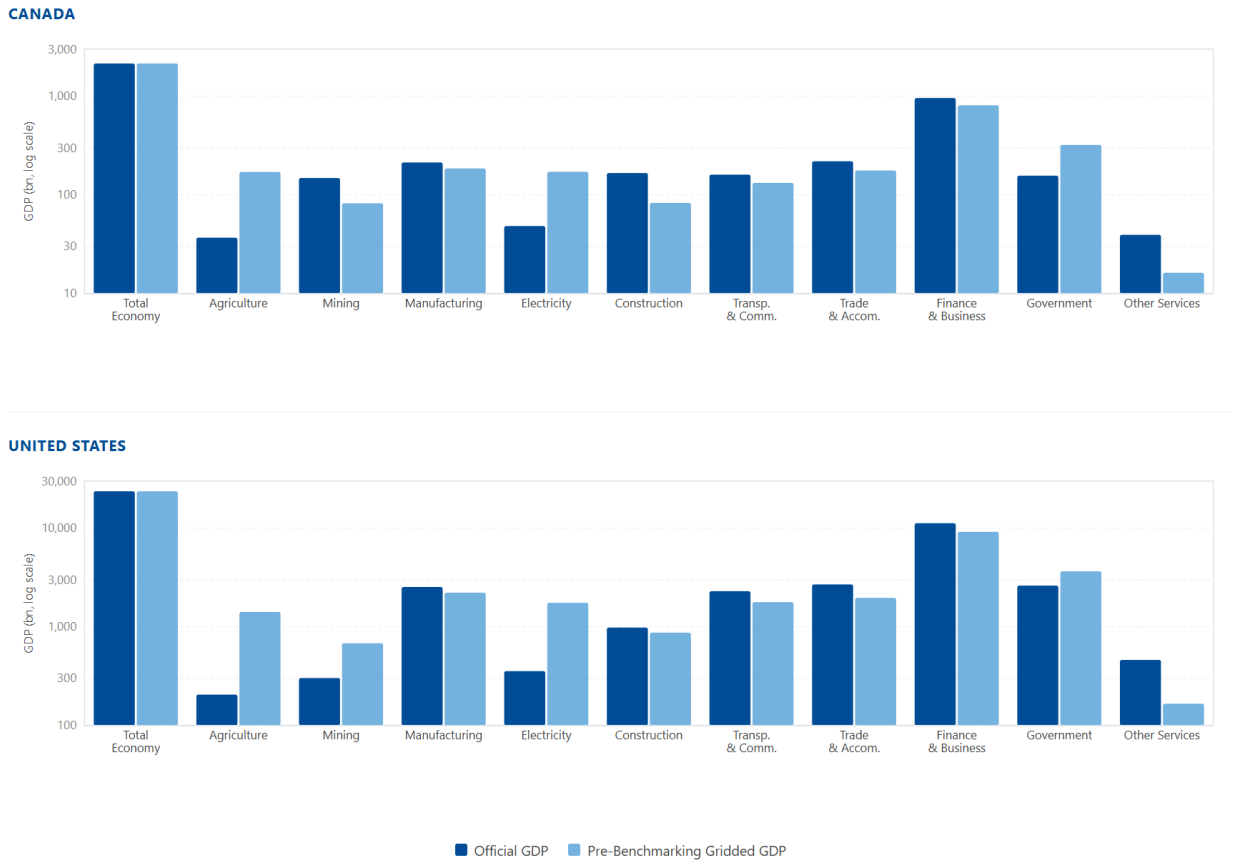
(c) Share of manufacturing



Source: Authors' computation.

Figure 7 reports the preliminary gridded GDP estimates compared with the official GDP by industry. The figure serves as a diagnostic of model fit before the benchmarking step is applied, revealing patterns in how the spatial random forest captures economic activity across sectors that inform future refinements to the methodology.

Figure 7: Preliminary Gridded GDP vs. Official GDP by Industry, Canada and United States



Source: Author's computations.

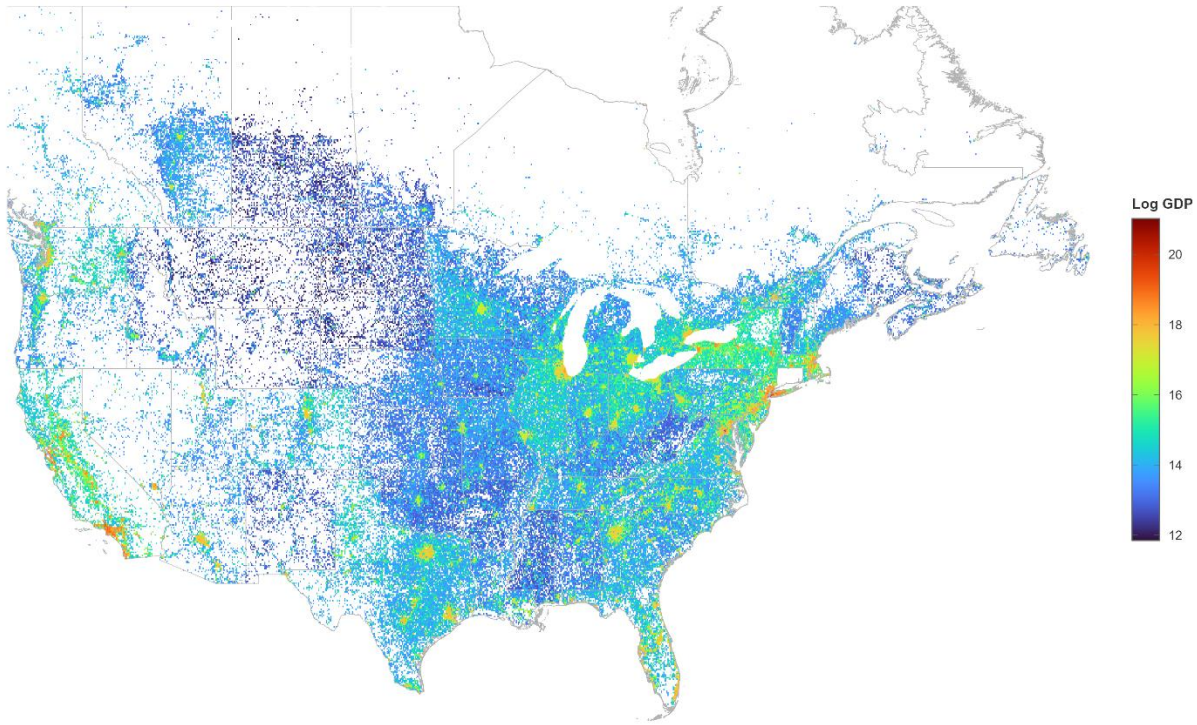
Agriculture and Utilities show the largest divergence, with pre-benchmarking gridded GDP estimates that are four to five times higher than the official data. These patterns are partly attributable to the current set of Earth observation covariates, which reflect land surface characteristics and nighttime light that do not always align closely with sectoral value added. Notably, the magnitude of this divergence is inconsistent with the in-sample fit reported in Table 2—Agriculture is among the better-performing sector models—indicating that the discrepancy cannot be attributed to predictive accuracy alone and warrants further investigation. Agricultural land, for instance, generates a large spatial footprint relative to its economic contribution, while public facilities are difficult to distinguish from higher-value commercial activity using land cover indicators alone. Industries with a strong urban and commercial presence — Financial Intermediation; Trade & Accom.; Manufacturing; and Transp. & Comm. — show closer alignment, with ratios broadly in the 75–90 percent range for both countries.

These discrepancies across all sectors are addressed through the reconciliation procedure, which scales gridded estimates to official national and subnational totals for each industry. Future iterations of the model will

incorporate industry-specific covariates better suited to capturing the spatial distribution of sectoral output, which is expected to reduce discrepancies in the machine learning estimates and improve the stability of the reconciliation step.

Figure 8 presents the post-reconciliation gridded GDP estimates for the total economy across Canada and the United States at 1 km² resolution. Each cell represents the spatially disaggregated share of official GDP allocated by the hierarchical reconciliation procedure, with red shading indicating higher economic output per cell. The logarithmic scale is used to accommodate the wide range of values across urban and rural areas while preserving visibility of spatial variation at lower density locations.

Figure 8: Gridded GDP, Total Economy, Canada and United States, 2021 (Post-Benchmarking, Log USD PPP)

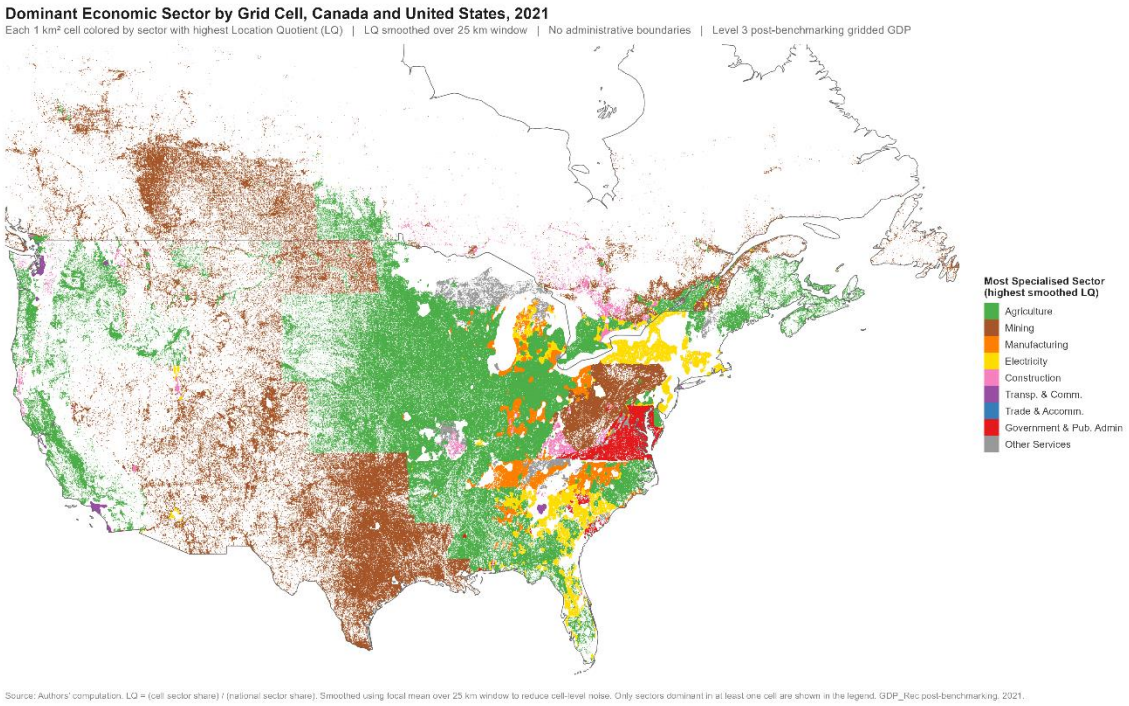


Source: Authors' computation.

Three patterns are worth noting. First, the map confirms that economic activity is highly concentrated in a small number of dense urban corridors: the highest output cells cluster in the major metropolitan areas of both countries, which account for the darkest shading on the map. Second, there is a clear east-west gradient in the United States, with near-continuous moderate activity across the eastern half of the country and sparse, low-value cells in the arid interior west, while the northern Great Plains display dispersed low-value cells consistent with extensive agricultural land use. Third, Canadian activity is visibly concentrated within a narrow band close to the U.S. border and disappears almost entirely at higher latitudes, reflecting the country's distinctive settlement geography.

Figure 9 presents the dominant economic sector at the grid-cell level across Canada and the United States in 2021, based on the Location Quotient (LQ)[10] of each sector relative to the national average. Each 1 km² cell is colored by the sector with the highest smoothed LQ, where values above 1.5 indicate above-average local specialization (e.g., when the share of sectoral value added in the cell is 50 percent larger than the national share.) The map is produced without administrative boundaries, allowing spatial patterns to emerge directly from the gridded data rather than being constrained by state or provincial borders.

Figure 9: Dominant Economic Sector by Grid Cell, Canada and United States, 2021



Source: Authors' computation.

[10] The LQ is the ratio of a sector's share of value added in a given grid cell to that sector's share of value added nationally, so that a value of 1.0 indicates a cell in which the sector is exactly as prominent as it is in the country as a whole.

Agriculture (green) dominates a vast contiguous corridor running through the interior plains of both countries, with additional concentrations in irrigated valleys and coastal lowlands. Mining, Oil and Gas (brown) forms distinct zones in the intermountain west, the Appalachian belt, and scattered extraction sites across northern Canada. Manufacturing (orange) traces the historical industrial belt around the southern Great Lakes and extends into the southeastern United States. The service-oriented sectors display a strongly urban pattern: Finance and Business Services (dark navy) concentrates in the largest metropolitan centers of the Northeast[11], while Government and Public Administration (red) forms a conspicuous cluster around Washington D.C. — one of the states with a dominant government sector. Utilities sector (yellow) appears in dispersed pockets, likely reflecting the location of large generation facilities relative to the sector's small national GDP share, which makes even moderate local activity produce high LQ values.

Two features of the map merit emphasis. First, sectoral specialization is a spatially continuous phenomenon that pays no attention to jurisdictional lines: the agricultural belt runs unbroken across the Canada–U.S. border, and the mining and manufacturing zones span multiple states and provinces. This continuity is invisible in administrative-level statistics and constitutes a distinctive analytical contribution of gridded GDP data. Second, the fine spatial resolution reveals within-state heterogeneity that administrative aggregates conceal, with different sectors dominating different areas of the same state or province (for example, mining dominance in western Pennsylvania coexisting with finance dominance around New York). For the U.S., this partly inherits official county data, while for Canada all within-province detail is model-generated.

Spatially downscaling GDP to the grid level carries important caveats. Economic activity is inherently interconnected across space: production draws on factors that move between regions — labor, capital, and intermediate goods — as well as on flows of intangibles such as know-how and intellectual property. Consumption, saving, and investment are similarly dispersed. Financial intermediation, electricity generation and transmission, and telecommunications are not spatially divisible in any straightforward way, since the location where value is created need not coincide with the location where the underlying physical infrastructure sits. National accounts resolve these flows at the macro level through established compilation rules, but those rules offer little guidance on how such activity should be allocated once the unit of analysis becomes a one-square-kilometer cell. Another important caveat to note is that model predictive performance varies significantly across ten-sector classifications. While the model performs well in detecting spatial patterns for overall economic activity such as GDP and sectors such as agriculture, mining, financial services, model must be retrained for other sectors such as public utilities. This could involve adding relevant geocovariates that reasonably capture spatial variation for specific sectors such as point of interest data.

Nonetheless, downscaling macro-level socioeconomic data is a well-established practice in research and policy analysis wherever granular data are unavailable. Modelled labor market estimates from the ILO, small-area estimates derived from population censuses, and subnational population projections are all widely used on this basis. Gridded GDP and industry data extend the same logic to economic activity, offering granularity that remains consistent with existing macroeconomic statistics. Applications include spatial heterogeneous-agent models, assessments of disaster impact on local economies, and studies of spatial spillovers — all of which require disaggregated data that can be reconciled with official aggregates.

[11] The ‘Finance and Business Services’ layer should not be read as the location of financial exposure or collateral: value added in finance, insurance, real estate and business services is often recorded at headquarters, which may diverge substantially from the location of the underlying real exposure.

VI. Conclusion

Gridded macroeconomic statistics are becoming increasingly important for economic analysis and policymaking. Official statistics are typically produced at the national or, at best, subnational level, which limits their ability to capture the spatial heterogeneity of economic activity. High-resolution economic data allow analysts to better understand the geographic distribution of production, identify regional disparities, assess exposure to physical and climate risks, and evaluate infrastructure and development gaps. By translating macroeconomic indicators into a spatially explicit format, gridded GDP provides a bridge between macroeconomic statistics and the rapidly expanding ecosystem of geospatial datasets used in climate science, disaster risk management, and development planning.

This paper contributes to this literature strand by introducing a methodology that combines satellite-based indicators, machine learning techniques, and official statistics to generate spatially detailed macroeconomic data. The approach integrates Earth observation datasets—including nighttime lights, built-up areas, land cover, and population grids—with official national and subnational GDP statistics through a hierarchical framework based on random-forest models. By anchoring machine-learning estimates to official macroeconomic aggregates, the method preserves consistency with national accounts while distributing economic activity across fine spatial grids. This combination of satellite observations, statistical learning, and official data provides a practical and scalable framework for producing gridded macroeconomic indicators.

The application to Canada and the United States demonstrates the potential of the methodology. The results show a high degree of consistency between model-based estimates and official subnational GDP figures, indicating that satellite-derived predictors and machine learning can effectively capture spatial patterns of economic activity. At the same time, the analysis highlights areas where improvements are needed. In particular, regions where economic activity is dominated by service sectors—often less visible in physical or remotely sensed indicators—present greater challenges for spatial allocation. Further work on sector-specific predictors and additional geospatial inputs could help improve accuracy in these contexts.

Looking ahead, several avenues exist to improve and extend the framework developed in this paper. On the estimation side, the spatial random forest models could be enriched with additional covariates better suited to capturing economic activity in rural and remote areas, where Earth observation proxies such as nighttime lights and land cover perform less reliably. Candidates include road network density, agricultural productivity indices, and distance to markets. Further refinement would involve incorporating population data as a consistency check on the spatial distribution of output, ensuring that benchmarked GDP is not allocated to grid cells with near-zero population where economic activity is implausible, thereby improving the face validity of the estimates in sparsely inhabited regions. Finally, incorporating information on the spatial distribution of service-related infrastructure and establishments, such as banks, government offices, schools, hospitals, and retail centers, could further improve the estimation of value added in service industries and enhance the overall fit of the model.

Geographically, the current coverage of Canada and the United States represents a proof of concept for advanced economies with well-developed official statistical systems; extending the methodology to other regions, including Latin America, Sub-Saharan Africa, and South and Southeast Asia, would significantly broaden the policy relevance of the framework, particularly for countries where subnational GDP data are sparse or published with long lags. Finally, the gridded GDP estimates developed in this paper are intended to serve as a core input to GeoPulse, a geospatial analytics platform currently under development at the IMF, which aims to integrate high-resolution economic, financial, and risk data to support macro-financial

surveillance and climate vulnerability assessments at the local level. Incorporating gridded GDP into GeoPulse would enable analysts to combine output estimates with spatially granular indicators of natural hazard exposure, financial stress, and demographic change, opening up a new generation of place-based economic analysis that goes beyond what conventional national accounts can support.

More broadly, this work underscores the importance for the IMF of incorporating granular data into macroeconomic analysis. As economic risks — including climate shocks, infrastructure disruptions, and regional inequalities — become increasingly spatial in nature, the ability to analyze macroeconomic indicators at finer geographic scales becomes essential. By combining official statistics with geospatial datasets and modern analytical tools such as machine learning, the IMF can enhance its capacity to monitor economic activity, assess vulnerabilities, and support more targeted policy responses for the membership.

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Appendix A: List of Data Sources & Classifications

Table A.1: List of Subnational Data Availability

	Level of Aggregation /Resolution	Zonal aggregation type	Temporal Coverage	Source
	Official Subnational Data			
Subnational GDP:				
United States	County (admin 2)		2001-2023	BEA
Canada	Province (admin 1)		1997-2021	StatCan
Ten-sectors industry:				
Unites States	County (admin 2)		2001-2023	BEA
Canada	Province (admin 1)		1997-2021	StatCan
	High-resolution covariates			
Population	3 arcsecond (~100 meters)	Sum		WorldPop
Nighttime light	3 arcsecond (~100 meters)	Sum	2015-2023	WorldPop
Distance to built surface	3 arcsecond (~100 meters)	Mean	2015-2030	WorldPop
Distance to highway	3 arcsecond (~100 meters)	Mean	2023	WorldPop
Average annual precipitation	3 arcsecond (~100 meters)	Mean	2015-2023	WorldPop
Average temperatures	3 arcsecond (~100 meters)	Mean	2015-2023	WorldPop
Distance to cultivated land	3 arcsecond (~100 meters)	Mean	2015-2023	WorldPop
Number of buildings	3 arcsecond (~100 meters)	Sum	2023	WorldPop
Building density	3 arcsecond (~100 meters)	Mean	2023	WorldPop
Building area	3 arcsecond (~100 meters)	Sum	2023	WorldPop
Land cover classification	3 arcsecond (~100 meters)	Mean	2015-2023	ESA
Annual average surface NO2	1km	Sum	1990-2020	NASA
Number of iron and steel plants within 50km distance	Number within 50 km radius of a grid	sum	Historical-2025	

Source: Authors' compilation.

Table A.2: Industry Classification and Paper Correspondence

The table below presents the correspondence between the 10-sector industry classification used in the Groningen Growth and Development Centre (GGDC) database and the industry categories used in this paper. The GGDC classification follows the International Standard Industrial Classification of All Economic Activities (ISIC), Revision 3.1.

#	GGDC Industry	ISIC Code	Paper Short Name	Paper Acronym
1	Agriculture, hunting, forestry and	AtB	Agriculture	AGR
2	Mining and quarrying	C	Mining	MIN
3	Manufacturing	D	Manufacturing	MAN
4	Electricity, gas and water supply	E	Utilities	PUB
5	Construction	F	Construction	CON
6	Wholesale and retail trade, hotels	GtH	Trade & Accom.	TRAD
7	Transport, storage and	I	Transp. & Comm.	TRAN
8	Finance, insurance, real estate	JtK	Finance & Business Serv.	FIN
9	Government services	LtN	Government & Pub. Admin	GOV
10	Community, social and personal	OtP	Other Services	OTH

Source: Groningen Growth and Development Centre (GGDC), 10-Sector Database. Authors' classification.

Appendix B: Reconciliation Method

Our machine-learning models are trained separately for GDP and industry shares in that grid-level predictions might not guarantee an aggregate consistent with official subnational and national totals. To address such issues, we implement a reconciliation technique that considers machine-learning (ML) predictions available at the grid–industry level within a given country and aligns them with official subnational (regional) and industry totals while preserving their spatial, sectoral, and (when relevant) temporal structure "at best", following the quadratic reconciliation framework of Di Fonzo and Marini (2011). In the following we describe the reconciliation approach.

Within a country, let:

- $g = 1, \dots, G$ denote grid cells.
- $i = 1, \dots, I$ denote industries.
- $r = 1, \dots, R$ denote subnational regions.

Let $p_{g,i}$ denote the preliminary ML estimate for grid g and industry i, and let $x_{g,i}$ denote the reconciled value. Stacking all $x_{g,i}$ into a vector $x \in \mathbb{R}^{GI}$, and similarly p, we impose consistency with official regional accounts.

Let $S \in \mathbb{R}^{R \times G}$denote the spatial aggregation matrix mapping grids into regions:

$$S_{rg} = \begin{cases} 1 & \text{if grid } g \in r \\ 0 & \text{otherwise.} \end{cases}$$

We consider two types of official constraints.

(i) Regional totals by industry

$$\sum_{g \in r} x_{g,i} = y_{r,i}^{\text{off}}.$$

In matrix form:

$$(S \otimes I_I)x = y_{r,i}^{\text{off}}.$$

(ii) Regional totals across industries

$$\sum_{i=1}^{I} \sum_{g \in r} x_{g,i} = y_r^{\text{off}}.$$

In matrix form:

$$(S \otimes 1_I')x = y_r^{\text{off}}.$$

Stacking all restrictions yields the compact system:

$$Hx = y_a,$$

where H is sparse and encodes the full regional–industry accounting structure within the country.

To preserve the ML spatial–industry pattern, we adopt a proportional quadratic loss:

$$\min_x \sum_{g,i} \left(\frac{x_{g,i} - p_{g,i}}{p_{g,i}} \right)^2$$

In matrix form:

$$\min_x (x - p)' W (x - p),$$

where

$$W = \text{diag}\left(\frac{1}{p_{g,i}^2} \right).$$

This criterion penalizes relative deviations, ensuring that spatial heterogeneity and industry composition implied by the ML model are preserved as much as possible.

.

We solve:

$$\min_x (x - p)' W (x - p)$$

$$\text{s.t.: } Hx = y_a.$$

The first-order conditions yield the generalized least squares (GLS) reconciliation estimator:

$$\boxed{x = p - W^{-1}H'(HW^{-1}H')^{-1}(Hp - y_a)}$$

The reconciled estimates:

1. Satisfy all official regional accounting constraints exactly;
2. Minimize weighted deviations from ML predictions;
3. Preserve relative spatial and sectoral structure within the country.

The correction term depends solely on the discrepancy $Hp - y_a$, i.e., the inconsistency between aggregated ML predictions and official regional totals.

The aggregation matrix H is highly sparse due to the local nature of spatial aggregation and the block structure induced by industries. Consequently, $HW^{-1}H'$remains sparse and can be efficiently inverted using sparse Cholesky or multifrontal factorization methods, making the approach scalable to very fine grid resolutions within each country. The reconciled estimator can be interpreted as a minimum-variance adjustment of ML predictions to official regional accounts. It integrates:

- Machine-learning spatial information,
- Industry accounting consistency, and
- Official subnational benchmarks.

within a unified quadratic optimization framework applied independently for each country.

Two features of the reconciliation procedure warrant clarification. First, the weight matrix $HW^{-1}H'$ is undefined for cells with a predicted value of zero. To ensure that W is well defined throughout, cells with no predicted activity are assigned a small notional value of 0.01 prior to reconciliation; this value is negligible relative to the scale of the preliminary series and is removed after the reconciliation step, restoring an exact zero to these cells. Second, because the generalized least squares solution is unconstrained, the procedure can in principle return negative reconciled values where the discrepancy between the preliminary grid and the administrative total is large. In the Canada-U.S. application reported here, the reconciliation did not produce negative values.

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Appendix C: Data Coverage

Table C-1 reports the number of grid cells and post-benchmarking GDP for all Canadian provinces/territories and U.S. states covered by the gridded GDP dataset.

Table C.1: Gridded GDP Coverage by Province and State, 2021				
Province / State	**Grid Cells**	**% National Cells**	**GDP (USD PPP bn)**	**% National GDP**
A. Canada				
Alberta	216,265	21.9	333.1	15.5
British Columbia	79,828	8.1	300.6	14.0
Manitoba	82,059	8.3	66.7	3.1
New Brunswick	35,835	3.6	33.9	1.6
Newfoundland and Labrador	22,389	2.3	32.5	1.5
Northwest Territories	1,705	0.2	4.2	0.2
Nova Scotia	33,071	3.3	42.4	2.0
Nunavut	670	0.1	3.9	0.2
Ontario	190,495	19.3	820.5	38.2
Prince Edward Island	6,588	0.7	7.1	0.3
Quebec	137,186	13.9	425.9	19.8
Saskatchewan	179,854	18.2	75.1	3.5
Yukon	2,716	0.3	3.2	0.1
Total Canada	**988,661**	**100.0**	**2,148.9**	**100.0**
B. United States				
Alabama	122,437	2.3	258.6	1.1
Alaska	13,995	0.3	57.3	0.2
Arizona	83,086	1.6	437.7	1.8
Arkansas	114,678	2.1	152.2	0.6
California	197,269	3.7	3,504.1	14.8
Colorado	123,113	2.3	450.4	1.9
Connecticut	16,775	0.3	299.2	1.3
Delaware	6,441	0.1	82.6	0.3
District of Columbia	269	0.0	158.7	0.7
Florida	107,370	2.0	1,296.5	5.5
Georgia	143,353	2.7	712.3	3.0
Hawaii	1	0.0	93.5	0.4
Idaho	67,913	1.3	98.5	0.4
Illinois	171,827	3.2	949.2	4.0
Indiana	115,766	2.2	429.2	1.8
Iowa	176,078	3.3	222.4	0.9

Table C.1: Gridded GDP Coverage by Province and State, 2021				
Province / State	Grid Cells	% National Cells	GDP (USD PPP bn)	% National GDP
Kansas	171,507	3.2	191.4	0.8
Kentucky	116,731	2.2	238.1	1.0
Louisiana	86,327	1.6	262.6	1.1
Maine	56,197	1.0	78.8	0.3
Maryland	32,525	0.6	451.3	1.9
Massachusetts	26,945	0.5	661.6	2.8
Michigan	165,720	3.1	587.5	2.5
Minnesota	211,247	3.9	418.9	1.8
Mississippi	110,486	2.1	128.0	0.5
Missouri	198,558	3.7	367.3	1.6
Montana	116,690	2.2	59.2	0.2
Nebraska	133,551	2.5	149.7	0.6
Nevada	34,913	0.7	202.4	0.9
New Hampshire	23,468	0.4	100.0	0.4
New Jersey	23,483	0.4	701.3	3.0
New Mexico	97,852	1.8	111.6	0.5
New York	140,389	2.6	1,924.0	8.1
North Carolina	134,118	2.5	664.7	2.8
North Dakota	122,699	2.3	62.1	0.3
Ohio	135,128	2.5	764.8	3.2
Oklahoma	163,320	3.0	213.0	0.9
Oregon	87,390	1.6	277.5	1.2
Pennsylvania	132,115	2.5	848.3	3.6
Rhode Island	3,902	0.1	67.4	0.3
South Carolina	78,675	1.5	271.9	1.1
South Dakota	111,944	2.1	61.6	0.3
Tennessee	113,109	2.1	443.3	1.9
Texas	462,199	8.6	2,075.8	8.8
Utah	65,020	1.2	233.3	1.0
Vermont	26,729	0.5	37.4	0.2
Virginia	113,020	2.1	627.3	2.6
Washington	108,233	2.0	701.1	3.0
West Virginia	63,654	1.2	84.0	0.4
Wisconsin	165,411	3.1	370.7	1.6
Wyoming	61,312	1.1	41.0	0.2
Total United States	**5,354,938**	**100.0**	**23,681.2**	**100.0**

Notes: Grid cells reported are those with positive GDP only; cells with no economic activity are excluded. Each grid cell covers approximately 1 km² at mid-latitude. GDP figures in current USD billions. % National Cells and % National GDP are computed relative to the respective country total.

Appendix D: Additional Figures

Figure D.1: Predicted Gridded Ten-sector Shares: Mining

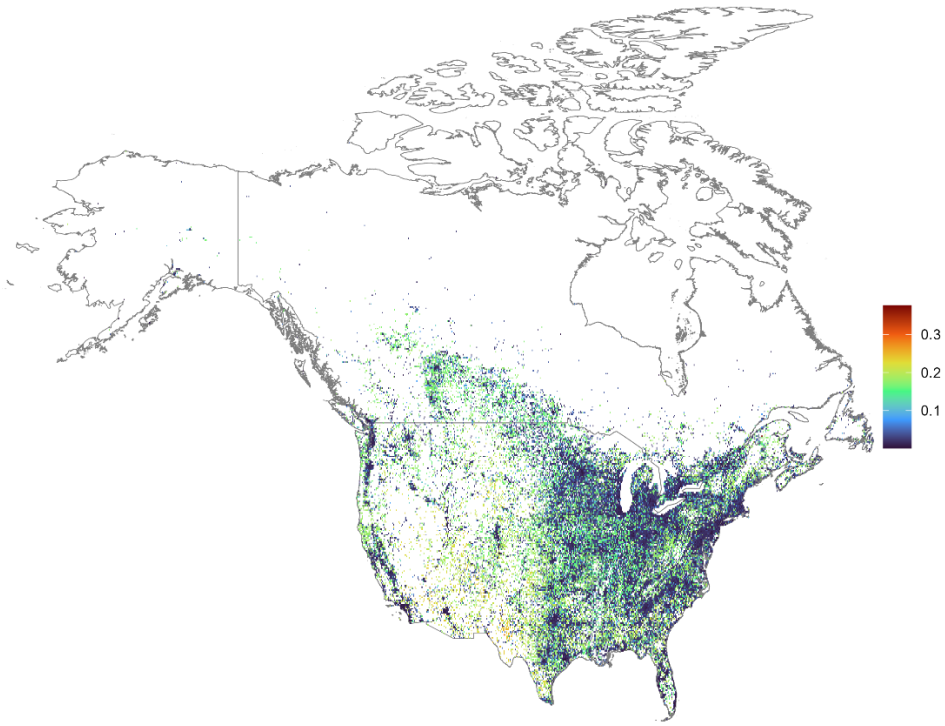


Figure D.2: Predicted Gridded Ten-sector Shares: Construction

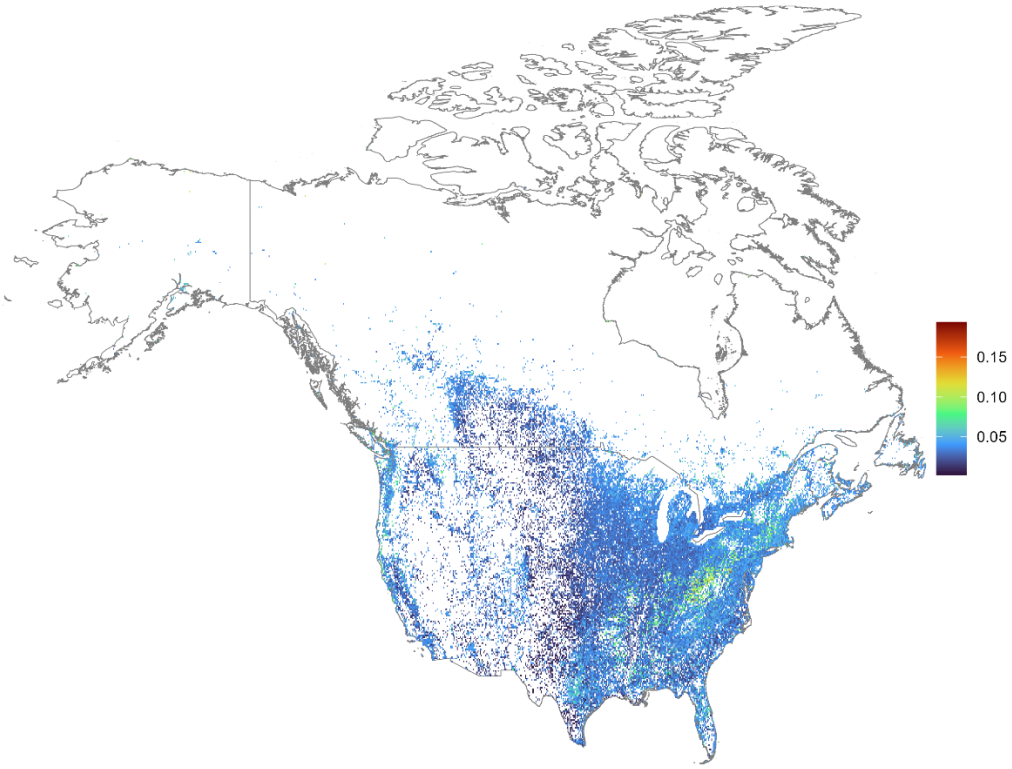


Figure D.3: Predicted Gridded Ten-sector Shares: Finance and Business Services

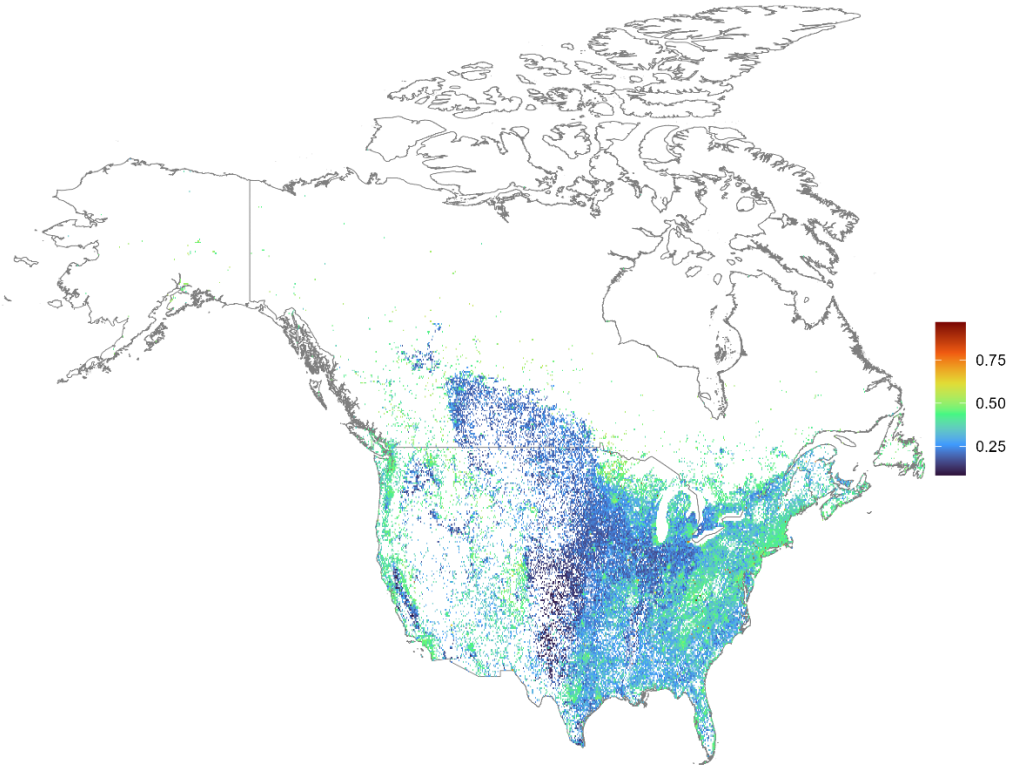


Figure D.4: Predicted Gridded Ten-sector Shares: Trade & Accommodation

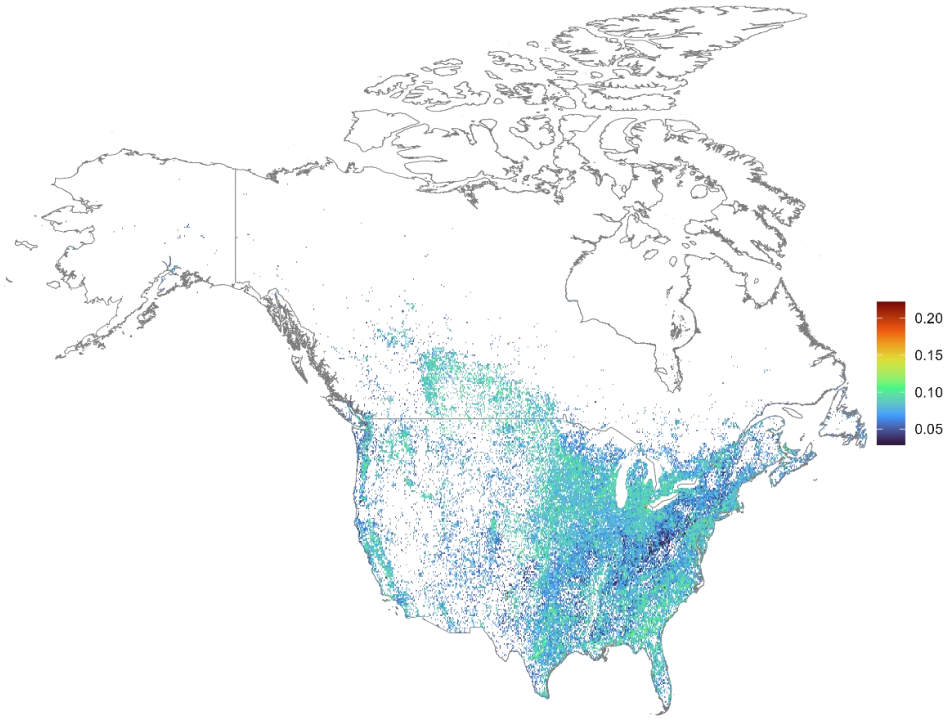


Figure D.5: Predicted Gridded Ten-sector Shares: Transport and Communication

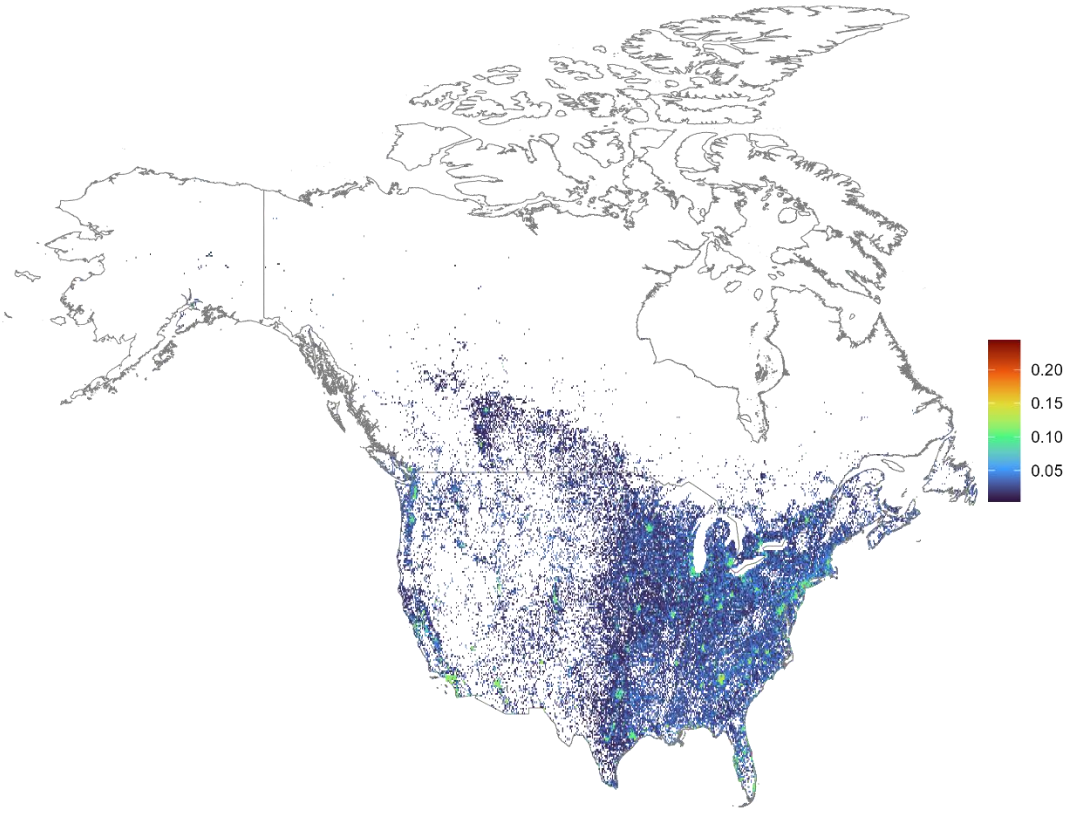


Figure D.6: Predicted Gridded Ten-sector Shares: Government & Public Administration

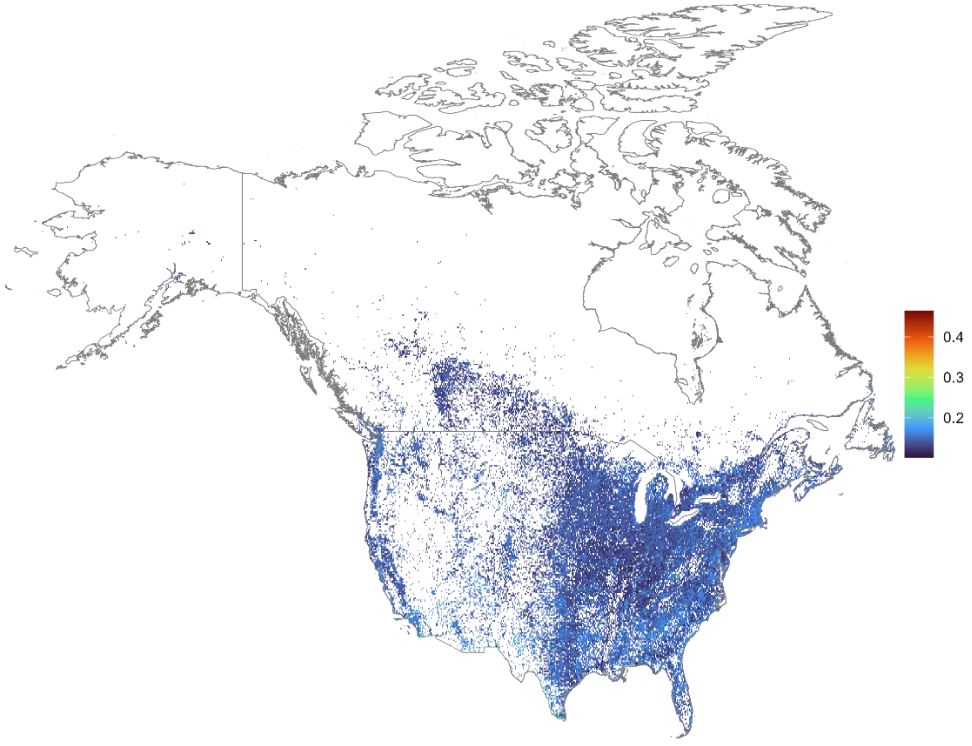


Figure D.7: Predicted Gridded Ten-sector Shares: Utilities

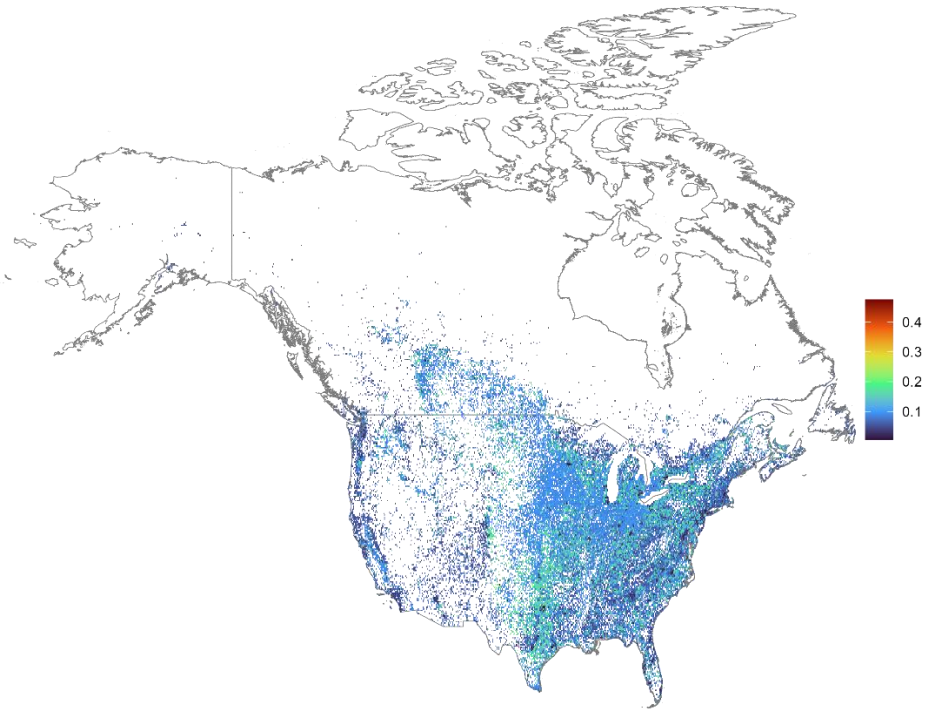
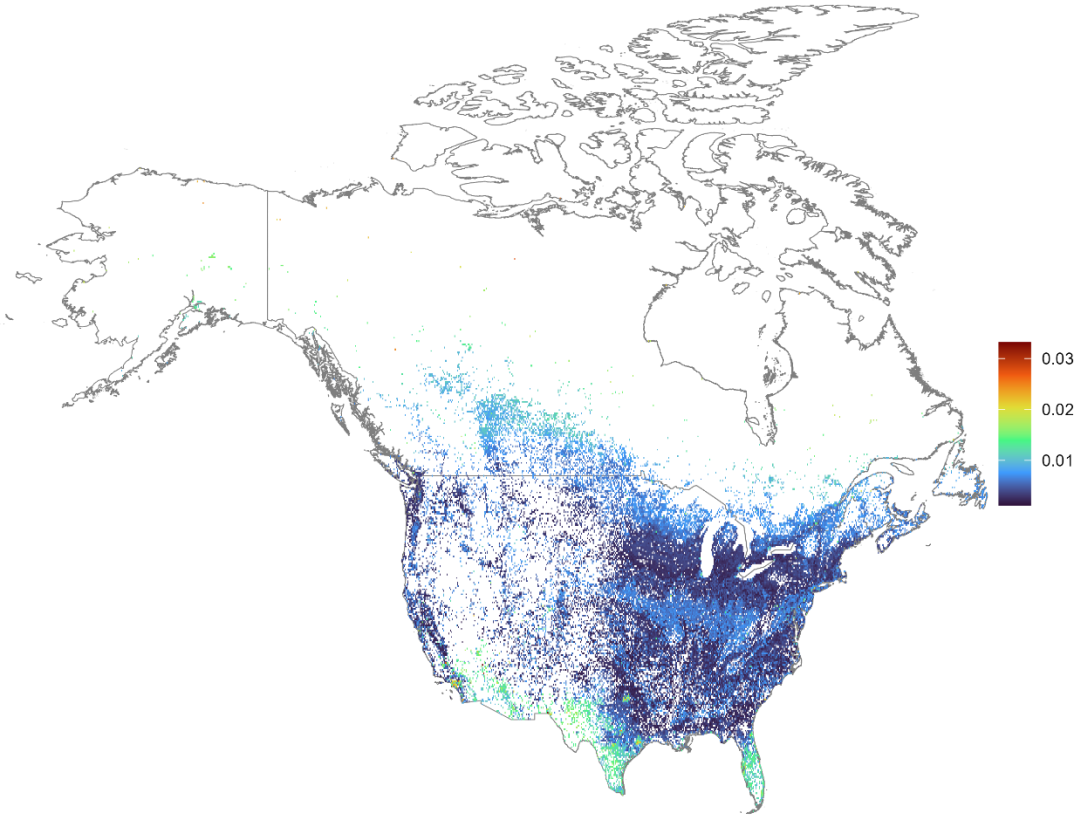


Figure D.8: Predicted Gridded Ten-sector Shares: Other Services





PUBLICATIONS

Earth Observations and Machine Learning for Gridded Macroeconomic Data

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